DEPARTMENT OF MATHEMATICS, IIT MADRAS

**MA5950 — Mathematical Finance: Theory & Implementation**

Instructor: Prof. Neelesh Upadhye

# Lecture 26: The Black–Scholes Formula

Solving the PDE · Interpreting $N(d_1)$ and $N(d_2)$ · Closing the Loop

October 11, 2026

# Today's Agenda

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3. What $N(d_1)$ and $N(d_2)$ mean
4. Put–call parity
5. Closing the loop: binomial $\to$ Black–Scholes

# Black–Scholes–Merton, 1973

**Key Takeaway**

$$C = S\,N(d_1) - Ke^{-r\tau}N(d_2), \qquad P = Ke^{-r\tau}N(-d_2) - S\,N(-d_1)$$

$$d_1 = \frac{\ln(S/K) + (r + \frac{\sigma^2}{2})\tau}{\sigma\sqrt{\tau}}, \qquad d_2 = d_1 - \sigma\sqrt{\tau}$$

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**Five inputs – note which is missing**

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# An Alternative Derivation

**The Risk-Neutral Recipe**

Under $Q$, the stock grows at $r$:

$$S_T = S_0 \exp\left[\left(r - \frac{\sigma^2}{2}\right) T + \sigma W_T^Q\right]$$

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Evaluating this Gaussian integral gives exactly the formula above.

# Everything From Week 5, Now Continuous

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# What the Two Terms Mean

**Key Takeaway**

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- $N(d_1) = \Delta$ – the **hedge ratio** from Lecture 25.

# Buy the Stock, But Only If You Have To

**The Option as a Conditional Purchase**

$Ke^{-r\tau}N(d_2)$: what you expect to *pay* – the strike, discounted, times the chance you pay it at all.

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# A Fully Worked Calculation

**$S$=100, $K$=105, $r$=5%, $\sigma$=20%, $T$=1**

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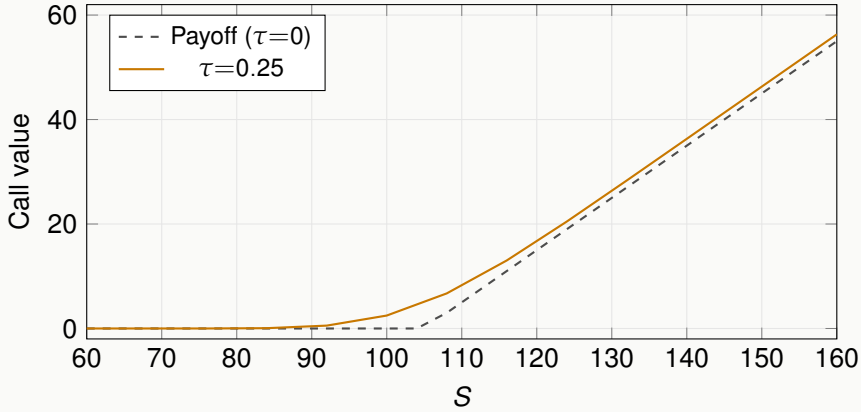
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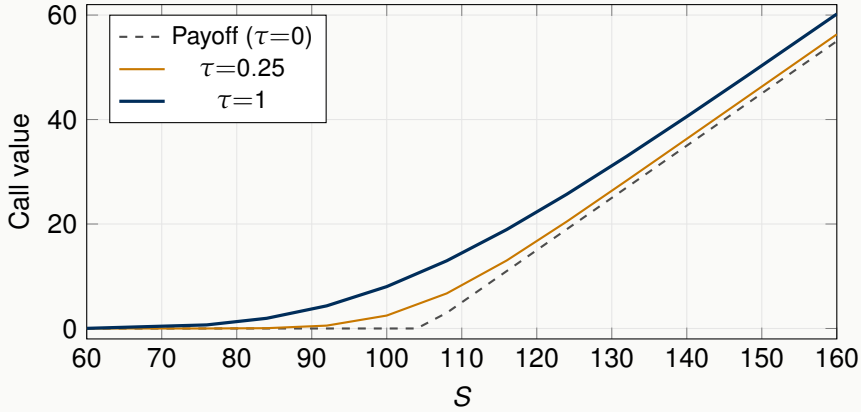
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$$P = \mathbf{7.9004}$$

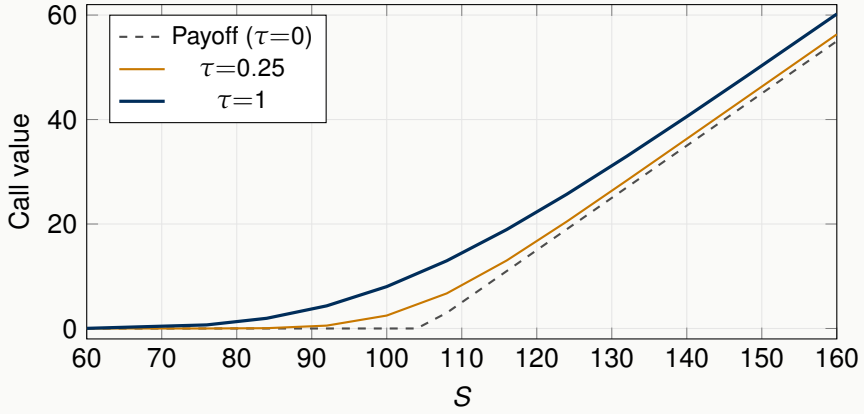
# Call Value vs. Stock Price



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The gap above the payoff is **time value** – largest near the strike.

# Put–Call Parity

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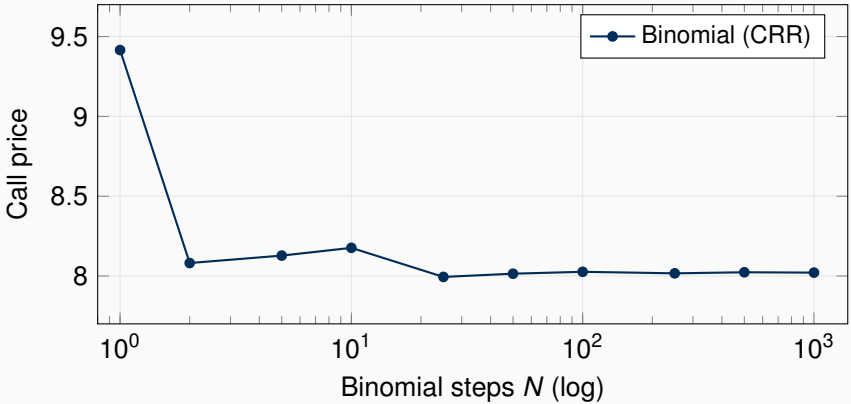
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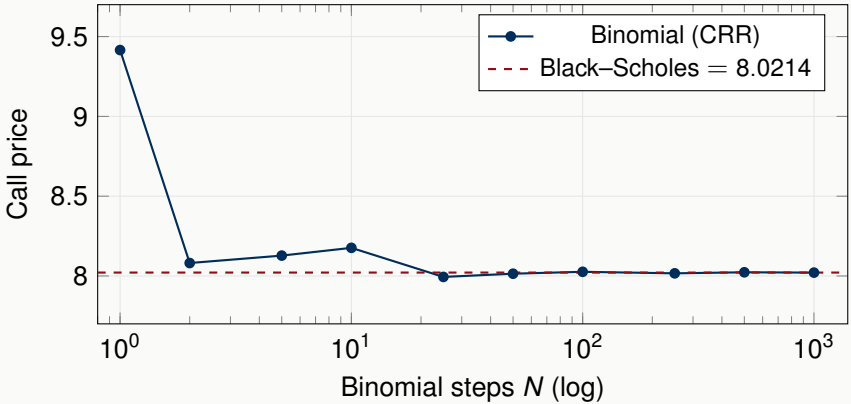
**Checked**

$C - P = 8.0214 - 7.9004 = 0.1209; \quad S - Ke^{-rT} = 100 - 99.8791 = 0.1209$ ✓

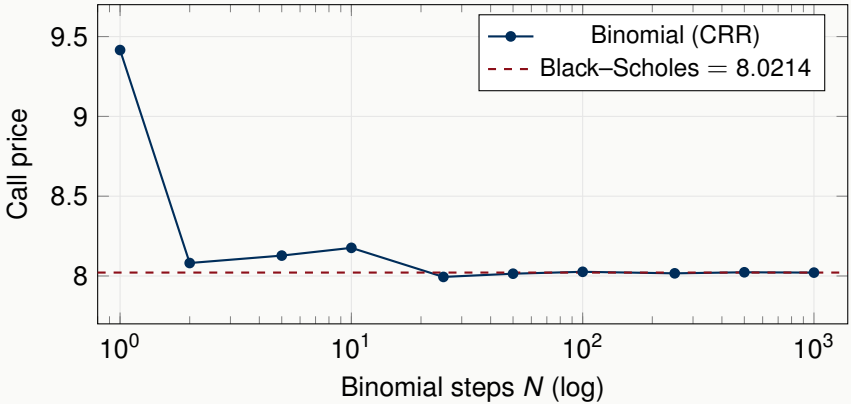
# The Week 6 Plot, With the Limit Named



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At $N = 1000$: 8.0211 vs. 8.0214 – agreement to four decimals.

# Two Roads to the Same Place

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Week 6: two states, a replicating portfolio, algebra on a tree.

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**Worth Pausing On**

Week 6: two states, a replicating portfolio, algebra on a tree. Today: stochastic calculus, a PDE, a Gaussian integral. Entirely different intellectual enterprises – landing on the **identical number to four decimals**. Not a coincidence: Donsker (Week 7) guarantees the discrete process converges, so the prices must too.

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- Parity: $C - P = S - Ke^{-r\tau}$.
- The exact limit of the binomial model.

# Check Your Understanding

## Discuss (2 minutes)

1. Price call and put for $S=80$, $K=75$, $r=4\%$, $\sigma=30\%$, $T=0.5$; check parity.

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2. If you sold 1000 such calls, how many shares to be delta-hedged?
3. As $\sigma \to 0$, show $C \to \max\left(S - Ke^{-r\tau}, 0\right)$. Why is that right?

# Looking Ahead

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Including how the market itself reveals the model's failure.